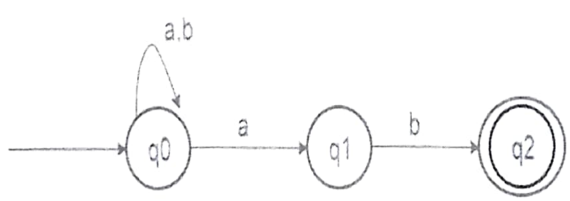
PAPER CODE: BCE-C612
SEMESTER EXAMINATION (MAY-2025)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: FORMAL LANGUAGES AND AUTOMATA THEORY

Time: 3 Hours **Max. Marks: 70**

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks. **(5 x 6 = 30 Marks)**

		CO	BL
Q-1	Define finite automata. Differentiate between DFA and NFA.	1	U
Q-2	Define Push Down Automata.	1	U
Q-3	Write a regular expression for a string having no consecutive zeros.Made of 0 and 1.	2	
Q-4	Define Melay and Moore machines.	1	U
Q-5	Prove 10+(1010)*[^+(1010)*]=10+(1010)*	3	K
Q-6	Draw a derivation tree for the input id+id*id for the following grammar: E→E+T/T T→T*F/F F→(E)/id	3	K
Q-7	Convert the following grammar into Greibach normal form. S→AA/a A→SS/b	4	AP
Q-8	Convert the following NFA to DFA:  	4	AP
Q-9	For the NFA in question no 08, find whether aaabaab is accepted or not.	1	U
Q-10	Explain 4 types of grammar.	1	U

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks.

(4 x 10 = Marks)

		CO	BL

Q-1 Consider the below finite automata and check the strings are accepted or not — 3 — AP

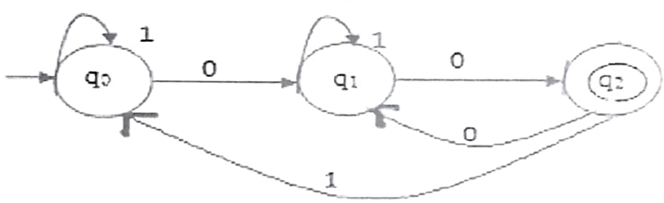
States (Q)	Input Alphabtes 0	Input Alphabtes 1
→q0	q1	q3
q1	q0	q2
q2	q3	q1
q3	q2	q0

(i) 1110 (ii) 0001 (iii) 1010

Q-2 Design a DFA equivalent for the given NFA; where q3 is the final state. — 4 — K

Current State	Input 'a'	Input 'b'
→ q0	q0, q1	q0 , q2
q1	--	q3
q2	q0, q3	q1
q3	q2	-

Q-3 Find the re for the following automata: — 3 — AP



Q-4 For the following PDA find the sequence of moves for input: aaabbb — 4 — K

$$\delta(q0, a, Z) = \{(q1, AZ)\}$$
$$\delta(q1, a, A) = \{(q1, AA)\}$$
$$\delta(q1, b, A) = \{(q2, \varepsilon)\}$$
$$\delta(q2, b, A) = \{(q2, \varepsilon)\}$$
$$\delta(q2, \varepsilon, Z) = \{(q3, Z)\}$$

Q-5 Construct a PDA which accepts strings made of a's and b's and it should be of type $a^n cb^n$, where n>=1. — 4 — AP

Q-6 Derive the string "aaabbabbba" for the following grammar: — 3 — K
S → aB | bA ; S → aS | bAA | a ; B → bS | aBB | b

Q-7 Define Turing Machine and also explain Universal Turing Machine — 1 — U

Q-8 Design CFG for $a^n b^n$ — 2 — E

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